



BICOL UNIVERSITY
COLLEGE OF SCIENCE
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Individual Journal

Name: Frederick B, Baldano

Group Name: Project Bloom

Role in the Group: Integration Specialist, QA Tester

Topic / Activity: Laboratory 4: REST API

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Overview of My Role

My role in the group focused on system integration and testing. I was responsible for ensuring that the API properly communicated with the database and that different client applications could successfully use the API. I also checked that the overall data flow, from request to response, was accurate and reliable.

Specific Tasks Performed

- Ensured that the connection between the API and the database is seamless using the configured connection file.
- Validated API endpoints such as `/login`, `/students`, and `/students/{id}/gwa` are functioning as intended.
- Tested protected routes to confirm that Bearer token authentication was properly enforced.
- Monitored the correctness of GWA computation and the corresponding academic remarks.
- Tested requests from different clients to confirm consistent API behavior.

Code Contributions

I handled the database connection setup in `db.php` using PDO to ensure stable and secure communication between the API and the database. This involved checking that the connection parameters were correct and that the API could consistently access and store data without errors. I also assisted in configuring authentication handling by verifying that Bearer tokens were properly validated in protected routes, ensuring that only authorized users could access sensitive endpoints such as student and subject records.

In addition, I reviewed and tested the request handling logic for different HTTP methods, including GET, POST, PUT, and DELETE, to confirm that each endpoint responded correctly based on the type of request. Part of my role was also to verify the implementation of business logic within the API, such as validating allowed grade values, preventing duplicate subject entries for the same student, and ensuring accurate GWA computation along with its corresponding academic remarks.

Challenges Encountered

One of the main challenges I faced was that some client requests were being blocked and could not retrieve data from the API. This was mainly caused by CORS restrictions and incorrect connection paths, which prevented proper communication between the client and the API. Another difficulty was my limited knowledge and experience in PHP, especially in handling API configurations and backend logic. Because of this, I had to spend additional time researching how PHP handles headers, routing, and database connections to ensure I could complete my tasks correctly.

I also encountered technical issues with the development environment. At times, XAMPP services would unexpectedly stop, which interrupted testing and delayed progress. There were also instances where MySQL Workbench and XAMPP conflicted due to overlapping ports, causing connection failures with the database. These issues made it harder to maintain a stable workflow during development and testing.

Problem-Solving and Contributions

To address the CORS issue, I added the necessary 'Access-Control-Allow-Origin: *' headers in the API, which allowed external clients to access the server without being blocked. I also reviewed and corrected the base URL and connection paths to ensure that all requests were properly routed to the API. To overcome my limited familiarity with PHP, I conducted research and reviewed documentation to better understand how API requests, headers, and database connections work, which helped me confidently apply the required fixes.

For the XAMPP-related issues, I ensured that all required services (Apache and MySQL) were properly running before testing and restarted them when unexpected stops occurred. I also resolved the port conflict between MySQL Workbench and XAMPP by checking and adjusting the ports used, ensuring that both could function without interfering with each other.